Wholesale Stock Management SaaS — Scope Draft

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What we're building

- Multi-tenant SaaS stock management for wholesalers (alcohol first; pharmacy and general later)
- Fast counter sales: customer says what they want, clerk keys it in, receipt prints on thermal printer
- Correct handling of nested packaging (box → six-pack → bottle; crate → 20/25 bottles)
- Fast product/price lookup, easy quotations
- Full stock management around it
- Backend in Axum, built step by step as the A–Z Axum learning project

Core design decision: packaging

- Stock stored in **one base unit** per product (e.g. bottle). Never store "2 crates + 5 bottles"
- Each product has **pack levels**: bottle = 1, six-pack = 6, box = 4 six-packs = 24, crate = 25
- Each pack level has: name, quantity in base units, own selling price, own barcode
- Sale line: "3 crates" → system converts to base units for stock, prices at crate price
- Stock display converts back: "1,203 bottles = 48 crates + 3 bottles"
- Break-bulk: opening a box is just selling at a lower level; no manual "unpack" step needed (optional audit record)

Modules

MVP

- Tenants, branches, users, roles/permissions
- Products, categories, pack levels, prices (per pack level, per price tier: retail/wholesale)

- Fast POS screen: search by name/code/barcode, keyboard-driven
- Sales + receipts (thermal 58/80 mm)
- Quotations → convert to sale
- Customers (incl. credit customers, balances)
- Payments: cash, mobile money, bank, credit; split payments
- Stock: purchases/receiving (GRN), adjustments, stock counts, low-stock alerts
- Suppliers
- Reports: daily sales, stock on hand, stock movement, profit, cashier shifts

Alcohol-specific

- **Returnable empties**: crate/bottle deposits charged and refunded, empties stock per customer
- Price changes by date (frequent supplier price changes)

Later

- Pharmacy: batch numbers, expiry dates, FEFO, controlled-drug records
- Multi-branch transfers
- Purchase orders
- Offline mode
- Mobile app

Thermal printing (needs decision)

- A browser web app cannot print raw ESC/POS directly to USB on Windows without workarounds
- **A. Small local print agent** (Rust, runs on the counter PC) — web app sends receipt to it; supports USB, Bluetooth and network printers. Most reliable.
- **B. Browser print dialog** (HTML receipt, printer driver installed) — simplest, slower, less control
- **C. Web Bluetooth / WebUSB** (Chrome only) — limited, fragile on Windows
- Recommendation: B for MVP, A as soon as stable

Open questions (answer before step 1)

1. Market: Tanzania, Kenya, or both? (affects fiscal receipts: TRA VFD/EFD vs KRA eTIMS — likely legally required)
2. Currency/VAT/excise: prices VAT-inclusive? Excise shown on receipt?
3. Returnable empties: in MVP? (strongly recommended for alcohol)
4. Frontend: Angular, or other?
5. Offline: does the counter need to keep selling when internet drops?
6. Barcode scanners in use?
7. Price tiers: one price per pack level, or different prices per customer type?
8. Credit sales: allowed? Limits per customer?
9. Test client: one shop or several branches?
10. SaaS billing model: per branch, per user, flat?